

Charging optimization in lithium-ion batteries based on temperature rise and charge time



Caiping Zhang^{a,b,*}, Jiuchun Jiang^{a,b}, Yang Gao^{a,b}, Weige Zhang^{a,b}, Qiujiang Liu^{a,b}, Xiaosong Hu^{c,d,*}

^a National Active Distribution Network Technology Research Center (NANTEC), Beijing Jiaotong University, Beijing 100044, China

^b Collaborative Innovation Center of Electric Vehicles in Beijing, Beijing Jiaotong University, Beijing 100044, China

^c The State Key Laboratory of Mechanical Transmissions, Chongqing University, Chongqing 400044, China

^d College of Automotive Engineering, Chongqing University, Chongqing 400044, China

HIGHLIGHTS

- Temperature rise and charge time constraint charging strategy is introduced.
- Genetic algorithm is wielded to optimize the charge current trajectories.
- The proposed method can reduce charge time notably with rational temperature rise.
- Battery charge performance and aging using the proposed approach are evaluated.

ARTICLE INFO

Article history:

Received 19 March 2016

Received in revised form 28 September 2016

Accepted 20 October 2016

Available online 2 November 2016

Keywords:

Polarization

Temperature rise

Charging optimization

Lithium-ion batteries

Electric vehicle

ABSTRACT

Lithium-ion battery fast charging issues have become a main bottleneck of large-scale deployment of electric vehicles. This paper develops a polarization based charging time and temperature rise optimization strategy for lithium-ion batteries. An enhanced thermal behavior model is introduced to improve the solution accuracy at high charging current, in which the relationship between polarization voltage and charge current is addressed. Genetic algorithm (GA) is employed to search for the optimal charging current trajectories. The effects of weighting coefficients of charging time and temperature rise on battery charging performance are discussed. The charging time of the optimized charging protocol is reduced by 50%, and the associated temperature rise is almost identical, compared to 1/3C constant current-constant voltage (CC-CV) charging. Aging experiments demonstrate that the proposed charging method has a similar capacity retention ratio to that of 0.5 CC-CV charging after 700 cycles, thereby accomplishing a good balance between charging speed and lifetime.

© 2016 Elsevier Ltd. All rights reserved.

1. Introduction

Electric vehicles (EVs) and plug-in hybrid electric vehicles (PHEVs) have been being rapidly developed in recent years, driven by the urgency to curb air pollutions caused by petroleum-dominant vehicles. The use of EVs and PHEVs will be increasingly widespread worldwide, and their annual sales will reach 100 million by 2050 [1]. Compared to internal combustion engine (ICE) fueled trivially; traction battery charging is much more complicated, due to its slow charging speed and unclear effects of charging

strategies on battery durability. Lithium-ion battery fast charging challenges constitute a principle bottleneck of EVs applications. Therefore, battery charging control is a thriving area of research in the field of EVs/PHEVs.

Numerous charging methods have been reported in the literature, with various objectives, e.g., increasing charging speed, enhancing charging performance, and maximizing battery life. Ref. [2] proposed a charging strategy of lithium batteries, based on an integration of Taguchi method and SOC estimation to search an optimal charging current profile. Guo et al. proposed an optimum charging technique for lithium ion batteries using a universal voltage protocol, which has the potential to improve charging efficiency and cycle life [3]. Ref. [4] developed a dual-objective optimal charging strategy based on equivalent circuit models, whereby charging time and charging energy loss are optimally traded off. Ref. [5] uses dynamic programming optimization

* Corresponding authors at: National Active Distribution Network Technology Research Center (NANTEC), Beijing Jiaotong University, Beijing 100044, China (C. Zhang). The State Key Laboratory of Mechanical Transmissions, Chongqing University, Chongqing 400044, China (X. Hu).

E-mail addresses: zhangcaiping@bjtu.edu.cn (C. Zhang), xiaosonghu@ieee.org (X. Hu).

algorithm to find a sub-optimal charging protocol under a certain balance between charging time and charging loss. Ref. [6] proposes an online tracking algorithm to allocate and track the optimal charging frequency for common batteries in real time under any condition. Ref. [7] presents a framework for optimizing lithium-ion battery charging subject to side reaction constraints, which can help avoid damage phenomena. Methods for battery charging optimization can be mainly categorized as improved charging current waveforms based methods [2,3,5,6,8–10], battery model based methods [4,7,11–16], polarization based methods [17,18], and enhanced battery material based methods [19]. Improved charging current waveforms based methods are generally simple to control and implement. However, they are often heuristic and lack of theoretical foundations in choosing battery charging current in an optimal manner. Battery model based methods can predict charging current by employing, e.g., a lumped equivalent circuit model, an ac-impedance model, or an electrochemical model. They combine external electrical behavior with internal reaction mechanisms, searching for optimal charging current. Diverse modeling and identification methods are developed to acquire parameters with high precision [20–24]. Nevertheless, effects of battery temperature and health fade on model parameters need to be further investigated. Polarization based methods provide acceptable charging current with constraints on battery polarization voltage. The polarization modeling and its quantitative effects on battery life are worth further examination. The enhanced battery material based methods improve the capability of battery charging acceptance by adding additives to the electrode materials or using new materials. However, stability and safety of such batteries need to be further examined, which cannot be applied in real-world EVs in a short term.

Temperature has a significant impact on lithium-ion battery performance and lifetime [25–27]. Battery activity increases as temperature is elevated. Nevertheless, if temperature increases over allowable limits, the stability of battery cathode lattice structure is getting worse, which not only accelerates battery degradation, but also results in battery safety hazards. Accelerated aging in thermal cycling for LMO batteries at 40 °C was observed, which are mainly caused by active material losses in both electrodes, generalized loss of lithium inventory, and inhibited kinetics [28]. During rapid charging, temperature gradient will inevitably increase since average charging current is enlarged, leading to lifetime decrease if operated in unreasonable thermal excursions. However, maximum charging currents differ at discrepant regions of battery state of charge (SOC), which makes balancing charging speed and temperature rise possible. In our previous study [18], an acceptable charging current curve has been pursued in accordance with lithium-ion battery polarization voltage behavior. The charging curve is able to prevent polarization from being out of range and thus be conducive to increasing charging capacity and charging speed.

This paper focuses on constraint analysis of temperature rise, optimal charging strategy, and reliable algorithm towards optimized charging patterns. There are three major contributions added to the related literature. First, an enhanced thermal behavior model is introduced to improve the solution accuracy at high charging current, and the relationship between polarization voltage and charge current is addressed. Second, a polarization based charging time and temperature rise optimization strategy for lithium-ion batteries is proposed to equilibrate charging speed and lifetime. Finally, genetic algorithm (GA) is wielded to search for the optimal charging current trajectories, taking temperature rise constraint and charging time into account. The implication of weighting coefficients is also explored.

The remainder of the paper is organized as follows. Section 2 describes the thermal model. Section 3 formulates the optimal charging strategy, reconciling temperature rise with charging time.

Section 4 presents experimental validations, followed by conclusions summarized in Section 5.

2. Enhanced thermal behavior model

Battery surface is assumed to have a uniform temperature when charge current is within some limits, which can be regarded as a particle cell for thermal modeling. The thermal behavior model can be expressed by the following equation [29]:

$$C_{\text{cell}} \frac{dT_{\text{cell}}}{dt} = Q_S + Q_O - Q_B \quad (1)$$

where C_{cell} represents heat capacity of battery, T_{cell} expresses battery temperature, Q_S is reversible reaction heat by entropy change ΔS , Q_O is the energy loss by the overpotential during charge and discharge cycles, and Q_B is the transferred heat. It should be noted that the average heat capacity is adopted in the study since it differs among active material components of battery. The temperature change of battery is governed by both endothermic transition and exothermic transition. The battery temperature is gradually rising while absorption of heat is greater than heat release due to heat accumulation.

The reversible reaction heat by entropy change can be described by

$$Q_S = T_{\text{cell}} \Delta S \frac{I}{nF} \quad (2)$$

$$\Delta S = - \frac{\partial \Delta G}{\partial T_{\text{cell}}} = nF \frac{\partial E}{\partial T_{\text{cell}}} \quad (3)$$

where F is Faraday constant, I is charge current, n is the number of moles of electrons transferred in the cell reaction ($n=1$ for a lithium-ion battery), ΔG is Gibbs free energy change for the cell reaction, and E is cell equilibrium that can also be replaced with close-to-equilibrium open-circuit voltage in approximate treatment. The differential $\partial E / (\partial T_{\text{cell}})$ is negative, and Q_S is endothermic during charging.

The transferred heat is expressed by

$$Q_B = hA(T_{\text{cell}} - T_{\text{amb}}) \quad (4)$$

where h is heat transfer coefficient, A is the total surface of cell, and T_{amb} is ambient temperature.

Cell overpotentials comprise mass transport polarization overpotential, charge transfer overpotential at positive and negative electrodes, and ohmic overpotential. It is usually for overpotential heat Q_O to be described as the square of current multiplied by overpotential resistance $Q_O = I^2 R_{\eta}$, in which the overpotential resistance R_{η} is regarded as constant [22], and the R_{η} at SOC = 0% is always applied. However, the polarization voltage has a linear relationship to charge current, rather than a proportional relation, and its slope is a function of SOC, which has been reported in our prior paper [12]. The simulation error of temperature using conventional calculation method will be consequently enlarged as the current increases. Considering the influence of SOC on polarization overpotential and ohmic overpotential, the overpotential heat can be rearranged as follows:

$$Q_O = Q_P + Q_J \quad (5)$$

$$Q_P = IU_P \quad (6)$$

$$Q_J = I^2 R_{\Omega} \quad (7)$$

where Q_P represents the energy loss by polarizations, U_P is cell polarization voltage expressed as $U_P = kl + b$ [16], and Q_J represents the energy loss by ohmic resistance.

Table 1
Sample battery parameters.

Capacity (Ah)	Weight (kg)	Total area of surface (cm ²)	Average heat capacity (J/g·K)	Charge limit voltage (V)	Discharge limit voltage (V)
25	0.899	753	0.9932	4.2	3.0

Discretizing Eqs. (1–7), by using the sampling time interval of 1 s, the battery temperature at time k can be expressed by

$$T_{\text{cell},k} = T_{\text{cell},k-1} + \frac{I^2 R_{\Omega} + IU_p + T_{\text{cell},k-1} I \frac{\partial E}{\partial T_{\text{cell}}} - Ah(T_{\text{cell},k-1} - T_{\text{amb}})}{mC_{\text{cell}}} \quad (8)$$

where m is the mass of battery. The initial temperature is 25 °C in this study.

Making use of the battery parameters shown in Table 1, the simulated and experimental results for battery temperature during charging are illustrated in Fig. 1, where R_{Ω} represents the simulated results with the DC resistance to calculate the overpotential heat, and $kl+b$ expresses the simulated charging temperature based on the improved thermal model. It is indicated that the simulation results of both methods are almost identical at the charge current of 1C, and the maximum error is approximately 0.9 °C. The simulation error with the $kl+b$ method is within 1.7 °C at the charge current of 2C. The accuracy is remarkably improved compared to the DC resistance calculation method with the maximum error of 2.7 °C. This is because that the polarization resistance is not constant, but decreasing to some extent, as the charge current increases. The estimation error with the DC resistance calculation method is small at the low charge current. Nevertheless, the constant DC resistance cannot effectively express battery polarization at high charge currents, resulting in larger errors. It is clear that the proposed $kl+b$ method can better describe the relationship between polarization voltage and current, demonstrating its high fidelity of thermal simulation, particularly in the case of fast charging.

3. Formulation of battery charging optimization

3.1. Theoretical analysis

The performance and lifetime will be significantly affected if the current flowing into the battery is beyond the maximum charge current value, even leading to safety issues. However, it has difficulty in obtaining the maximum charging current in real time from the aspect of electrochemistry, since it requires battery chemical parameters and detailed knowledge of battery material properties normally unavailable to BMS designers. The polarization voltage describes the deviation of the battery terminal voltage from its equilibrium, which can be observed based on lumped equivalent circuit models. The side reactions and lithium deposition may occur during charging when the polarization of the battery is too large, noticeably affecting battery performance and life. It corresponds to acceptable charging current of battery to a large extent. The polarization constraint is therefore employed to calculate the maximum charging current in this work. The charging currents at various SOC values are optimized within the charging boundary curve. The acceptable charge current is high at the low side of battery SOC, and becomes declining with augmented SOC, based on polarization properties [18]. The maximum charge current is determined by battery kinetics limits. The temperature rise needs also to be considered to extend battery lifetime when designing charging patterns. In consideration of battery charge polarization and

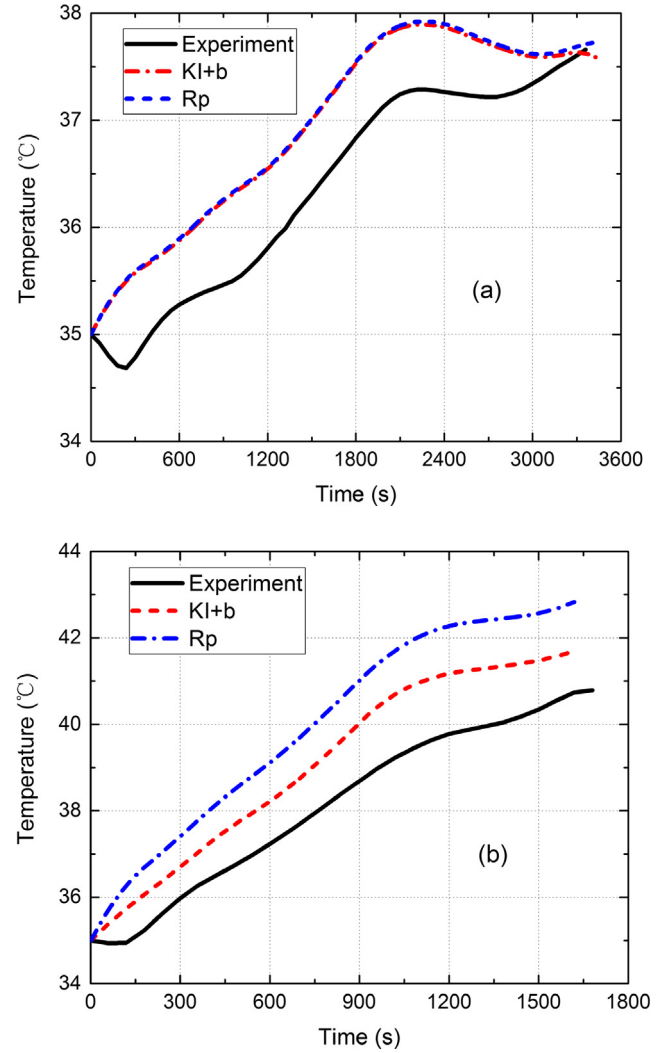


Fig. 1. Simulated and experimental temperatures during battery charge: (a) 1C and (b) 2C.

temperature rise constraints, the optimized charging strategy can be summarized in the following aspects. First, taking the acceptable charge current as the optimal charge current limit, the battery is charged with high current at the initial charging stage to speed up the charging process. Smaller charge current is then employed at the end of charging to decrease battery polarizations, so as to procure more charge capacities. Meanwhile, the temperature rise is calculated as a constraint, preventing thermal runaway and ensuring charge safety. Because the ohmic resistance, polarization resistance, and entropy change coefficient of the battery vary with SOC, the temperature rise rate differs among various SOC regions. Second, considering temperature rise characteristics at different SOC regions, the charge current can be augmented in SOC regions with lower temperature rise. The charge current and temperature rise can consequently be optimized at the whole SOC range, balancing battery lifetime and charging speed.

The whole SOC ranging from 0% to 100% is divided into N segments in the proposed charging strategy, and the charge current for each step needs to be optimized based on the charging time and temperature rise constraints. The SOC change for each segment is defined as ΔSOC , the relationship between N and SOC is expressed as $N = \frac{100}{\Delta\text{SOC}}$. The charging time for the k^{th} step can be given as

$$t_k = \frac{Q_k}{I_k} = \frac{\Delta SOC \cdot Q}{I_k} \quad (9)$$

where I_k is the charging current for the k^{th} step, Q_k represents the charged capacity, Q is the maximum available capacity of the battery, charged at a current of 0.05C. Then the total charging time can be calculated by

$$t = \sum_{k=1}^{N-1} t_k + t_N = \sum_{k=1}^{N-1} t_k + \frac{Q_N}{I_N}, \quad (10)$$

and the total charged capacity can be derived by

$$Q_{\text{ch}} = \Delta SOC \cdot Q \cdot (N-1) + Q_N. \quad (11)$$

Here the charging time and charged capacity for the N^{th} step are calculated alone, since the Q_N may not reach the theoretical value $\Delta SOC \cdot Q$ resulting from the charging current higher than 0.05C, and the total charged capacity is consequently less than the maximum available capacity. Supposing the charging voltage limit is 4.2 V, the open circuit voltage (OCV) at the end of charging for the N^{th} step can be described as

$$OCV_N = 4.2 - U_{\Omega,N} - U_{P,N} = 4.2 - (I_N R + k I_N + b) \quad (12)$$

Based on SOC-OCV mapping curve, the SOC at the end of charging for the N^{th} step can be expressed by

$$SOC_N = f^{-1}(4.2 - I_N R - k I_N - b). \quad (13)$$

The charged capacity Q_N is given by

$$Q_N = Q \cdot [SOC_N - \Delta SOC \cdot (N-1)]. \quad (14)$$

The charging time t_N can be obtained based on the formula $t_N = Q_N / I_N$. Using the proposed thermal model, the charge temperature rise for the k^{th} step is:

$$m C_{\text{cell}} (T_k^{\text{end}} - T_k^{\text{st}}) = \int_0^{t_k} I_k^2 R dt + \int_0^{t_k} I_k U_P dt + \int_0^{t_k} T_{\text{cell}} I_k \frac{\partial E}{\partial T_{\text{cell}}} dt - \int_0^{t_k} A h (T_{\text{cell}} - T_{\text{amb}}) dt \quad (15)$$

$$\Delta T_{\text{max}} = T_{\text{max}} - T_1^{\text{st}} \quad (16)$$

where T_{max} is the maximum temperature for the entire charging process.

The total charging time and temperature rise are two parameters to be optimized based on the proposed charging strategy. The objective is to speed up the charging process and to reduce

the resultant temperature rise, hence striking a balance between such two parameters. The fitness function is consequently portrayed by the following equation:

$$\begin{cases} F(t, \Delta T) = \alpha \left[\frac{40}{t_{\text{max}} - t_{0.05C}} (t - t_{0.05C}) + 60 \right] + \beta \left[100 + \frac{40(\Delta T - \Delta T_{0.05C})}{\Delta T_{0.05C} - \Delta T_{\text{max}}} \right] \\ \alpha + \beta = 1 \end{cases} \quad (17)$$

where α and β are weighting coefficients of charging time and temperature rise, respectively. Moreover, t_{max} represents the total charging time when the battery is charged with the acceptable current, whereas $t_{0.05C}$ is the charging time at the current of 0.05C. Here, $\Delta T_{0.05C}$ is the battery temperature rise when charged at the current of 0.05C. The linear weighting method is used to calculate the fitness values.

3.2. Optimal charging current calculation by genetic algorithm

Genetic algorithm (GA) is an adaptive optimization technique, including selection, crossover, and mutation which is inspired by the principles of natural selection and survival of the fittest from biological evolution. GA is an efficient method for solving both constrained and unconstrained optimization problems. GA requires a genetic representation of the solution which is so-called the population and a fitness function used to evaluate the solution. To use GA find the optimal solution, first randomly initialize a population of solutions and then through repetitive application of the selection, crossover, and mutation operators to produce the next generation, which make the population evolves towards an optimal solution. This heuristic (also sometimes called a metaheuristic) is routinely used to generate useful solutions to optimization problems in engineering [30,31]. The flowchart of optimal charge current estimation using the GA method is shown in Fig. 2, and the specific procedure is summarized as follows. An initial population is first produced as the basic currents. In the process, m column vectors with N charging currents are randomly generated based on the acceptable charge current limits. Second, fitness values are calculated, in which the charging time and temperature rise are regarded as the optimization objectives. And then the selection operator, recombination operator, and mutation operator are carried out in the population to optimize the parameters. Finally, the procedure is ended when meeting preset termination criterion. Actually each SOC segment has m optional charge currents, thus will have m^N charging current permutations for the whole charging

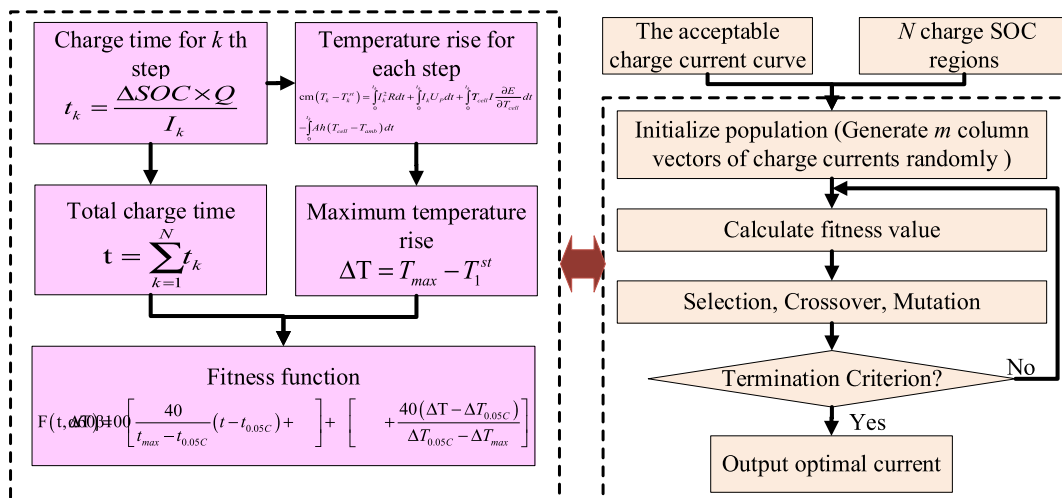


Fig. 2. Flowchart of charge current estimation with GA method.

process. Each permutation has a fitness values, and the charging current permutation with the highest score will be finally selected.

4. Results and discussion

4.1. Effect of weighting coefficients

The simulated charging current at various weighting coefficients is illustrated in Fig. 3, where α and β represent weighting coefficients of charging time and temperature rise, respectively. It is shown that the maximum fitness value of every generation approximately trends to be stable when the reproduction genera-

tions of the GA approach approximately reach 40, indicating that the fitness function is convergent. The overall variation tendency of the optimal charging current at various weighting coefficients is decreasing gradually as the acceptable charge current is declining with SOC increase. However, in some SOC regions, the charging current increases resulting from smaller resistance. According to the simulation results, when $\alpha : \beta = 3 : 7$, the total charging time is 119 min, and the temperature rise is 1.9 °C; when $\alpha : \beta = 7 : 3$, the charging time of the optimal current is 64 min, and the temperature rise is 3.8 °C. Considering various weighting coefficients, the optimization results sufficiently reveal interactions between charging time and temperature rises. Charging time will decrease with a high value of α and a small value of β . The relationship

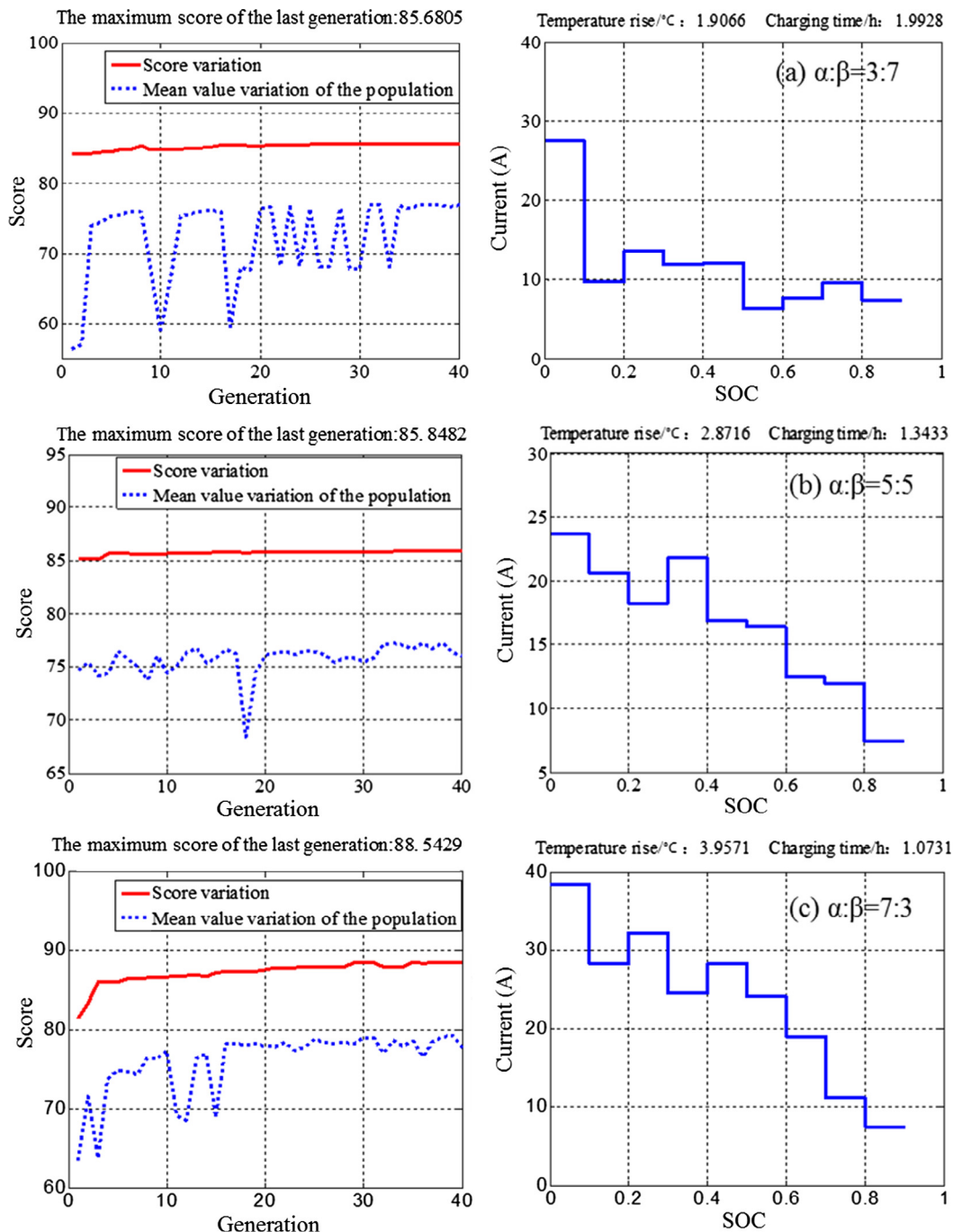


Fig. 3. Optimization results of charging current at various weighting coefficients: (a) $\alpha : \beta = 3 : 7$, (b) $\alpha : \beta = 5 : 5$, and (c) $\alpha : \beta = 7 : 3$.

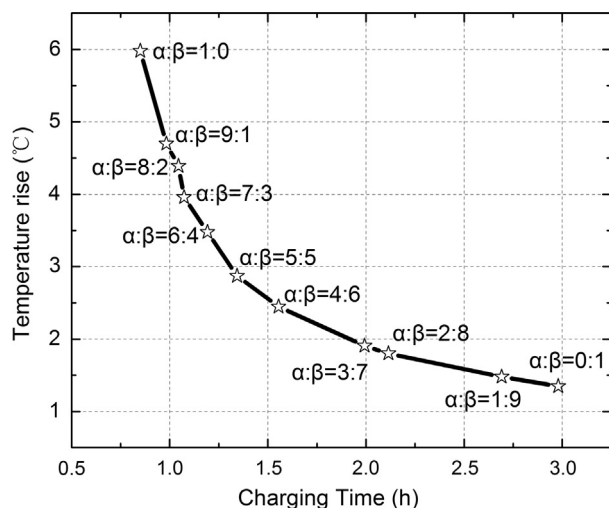


Fig. 4. Evolution of charging time and temperature rise, as the ratio of α to β decreases.

among charge time, temperature rise, and weighting values is shown in Fig. 4. It can be seen that charging time exhibits a monotonically increasing function of the ratio of α and β , while temperature rise is monotonically decreasing, without having a local optimal solution. To speed up the charging process and to minimize battery life degradation, the ratio value of α to β , 7:3, is finally picked.

4.2. Experiment results

A series of battery charging experiments were conducted to validate the proposed charging strategy. The battery testing platform is shown in Fig. 5. The instrumentations used to perform the experiments in this paper embrace Arbin Testing System, thermostat,

and PC. An Arbin Instrument BT2000 battery cycler was used to carry out the charge and discharge test. The volume of the thermostat is about 0.5 m³. The measured temperature range is 233.15–373.15 K. The battery used in this study is LiNMC cathode and graphite anode lithium ion batteries. Its parameters have been illustrated in Table 1. The battery was placed in the thermostat to keep the battery at desired temperatures.

Experiments involved the proposed charging protocol, a CC-CV charging whose current rate is an average in the former (average charge rate CC-CV), and a 1/3C CC-CV charging. The average charge rate is calculated by dividing the charge capacity by charging time. The experiment outcome is displayed in Table 2. It can be found that the proposed charging solution is 5 min faster than the average charge rate CC-CV charging, with almost the same total charge capacity and temperature rise. This is because more attention is paid to improving charging speed in our protocol. Specifically, CV charging time in our protocol is almost 60% less than that of the average charge rate CC-CV charging. The capacity during the CV part in the proposed charging only accounts for 2.4% of the total charge capacity, whereas that in the average charge rate CC-CV charging is 7%. Smaller charge current is then employed at the end of charging in the proposed charging to decrease battery polarization, so that its polarization at the end of charging is smaller than that of the average charge rate CC-CV charging, as confirmed in Fig. 6. As such a lower polarization level results in more charge capacity until 4.2 V, and less CV charge time and capacity, the CV charging can be neglected in the proposed charging protocol to reduce the total charging time. Compared to the 1/3C CC-CV charging, the charge time of the proposed solution when $\alpha:\beta = 7:3$ is reduced by about 60%, and the temperature rise is comparable, underlining the efficacy of our charging scheme.

From the experimental results above, it can be found that charge time and temperature rise can be optimized at the whole SOC region using the proposed method. The polarization is decreased during the charging process, and more capacity can be charged, desirably balancing charge capacity and charge time.

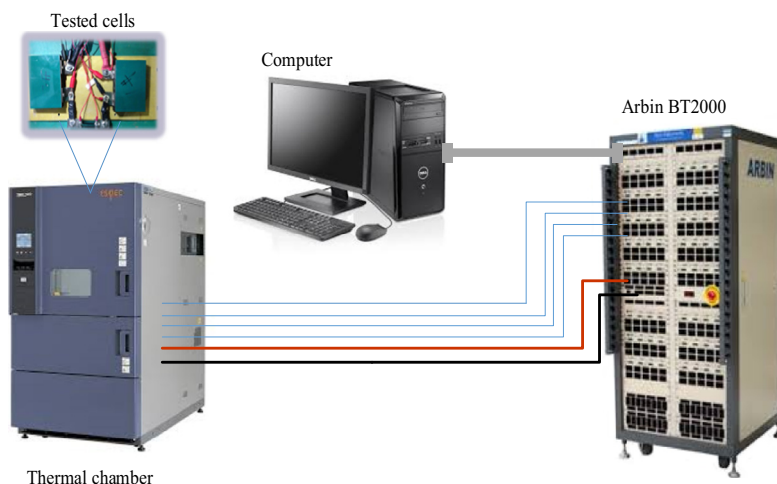


Fig. 5. Battery testing platform.

Table 2
Experimental comparisons of the proposed and conventional charging patterns ($\alpha:\beta = 7:3$).

Charging pattern	Total charge time (s)	Total charge capacity (Ah)	Charge time to 4.2 V (s)	Charge capacity to 4.2 V (Ah)	Temperature rise (°C)
Proposed	3938	19.912	3520	19.428	3.8
Average charge rate CC-CV	4263	20.084	3219	18.305	3.7
1/3C CC-CV	10,089	20.661	8997	18.742	1

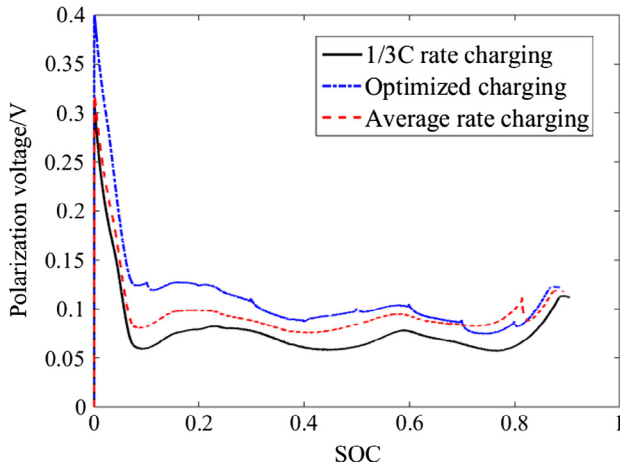


Fig. 6. Polarization voltage characteristics at different charging strategies.

With low temperature rise and polarization level, the proposed charging protocol enables alleviation of battery degradation as well.

4.3. Aging mechanism comparisons of two charging protocols

To further validate the capacity degradation performance of the proposed optimal charging, two cycling experiments were conducted. In the first experiment, the sample battery I was subjected to cycling with the widely used 0.5C CC-CV charging method and discharge at the current of 0.5C until the battery terminal voltage dropped to 3.0 V. In the second experiment, the sample battery II was subjected to cycling with the optimal charging protocol (α : β = 7:3) and discharge in the same way as the first experiment. Every 100 or 200 cycle times, the two sample batteries had a reference performance test (RPT) with charge-discharge cycles at C/20, C/2, 1C, 1.5C, 2C and 2.5C, to establish their performance characteristics. As such, the aging mechanisms of the two charging protocols could be evaluated. Also, the optimal charging protocol was adapted to the performance changes owing to battery degradation. All of the experiments were carried out at an ambient temperature of 25 °C.

Fig. 7 presents the evolutions of capacity for the two cycle aging regimes, where the capacity was determined in the RPTs by C/20 discharge. The discharged capacity is normalized by the initial 1/20C discharged capacity. The capacity retention ratio of the

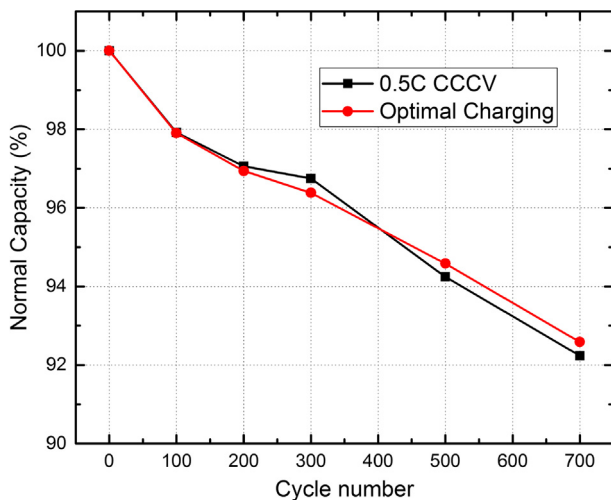


Fig. 7. Capacity variations of the two cycle aging regimes measured by reference performance tests (RPTs) using C/20 discharge.

optimal charging after 700 cycles is 92.59%, in contrast to 92.23% capacity retention ratio with 0.5C CC-CV charging. The capacity fade of the two charging regimes is almost the same, whereas the charging time of the optimal charging is only half of that of the 0.5C CC-CV charging. This implies that the optimal charging can significantly improve charging speed, while producing a negligible side effect on battery health. The proposed optimal charging method can thus balance charging time and charging life very well.

Incremental capacity analysis (ICA) has been recognized as an efficient method to diagnose battery degradation. ICA is leveraged to identify degradation mechanisms of the two batteries in the two cycle aging regimes [28]. Fig. 9 illustrates the evolutions of the IC curves under the two cycle aging regimes, where the IC curves were determined by the C/20 charge regime at cycle 0, 100, 200, 300, 500, and 700. The IC curves in Fig. 9 shows the IC peak indexed by convoluting positive electrode (PE) reactions (①, ②) with the five negative electrode (NE) reactions, noted as (①–⑤). The detailed descriptions of the characters ①, ②, ⑤ and the characters ①, ② are reported in Ref. [33]. Some IC peak may be invisible in the curves. The IC signature of the batteries experience several evolutions along with cycle aging. Some of the IC peaks can be exploited to signify electrodes' aging status. For example, the evolution of peak intensity associated with peaks ① * ① and ② * ② with cycle number can be used to identify degradation mechanisms [32].

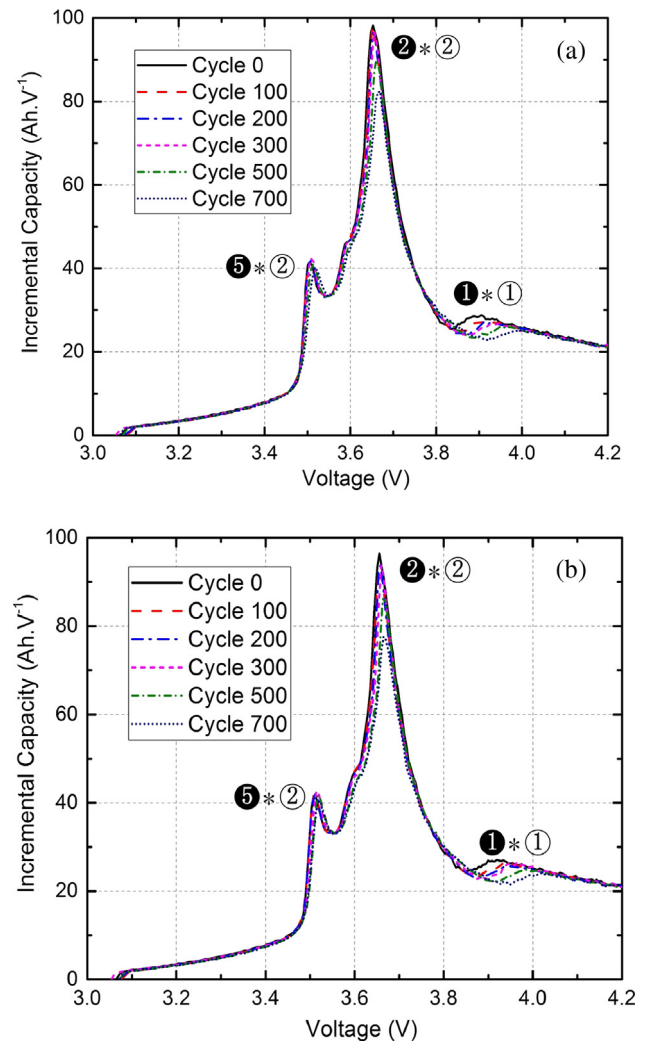


Fig. 8. Evolutions of the IC curves calibrated in the C/20 charge regime at cycle 0, 100, 200, 300, 500 and 700: (a) 0.5C CC-CV cycle aging and (b) optimal charging cycle aging.

For the battery I (0.5C CC-CV cycle aging), the intensity of peak ① * ① decreases noticeably, accompanied by peak position moving to high potential, since the aging cycle started. While the intensity of peak ② * ② seems to be unaltered before aging cycle 300, and between aging cycle 300 and 700, the intensity decreases obviously. For the peak ③ * ②, the intensity decreases only after aging cycle 500. These observations convey that the amount of active material involved in the reaction of NE and PE has not been compromised because full capacity of reaction ② is attained between aging cycle 0 and 300. In spite of evolutions in peak intensity, the shape of peaks ② * ② and ③ * ② remain almost intact, unveiling that the battery reaction kinetics have not been damaged in an apparent manner, so that the capacity fade should not be put down to kinetics degradation or impedance increases. Between aging cycle 0 and 300, the intensity decreasing of peak ① * ① suggests that a less amount of Li participates in the reaction ①. Loss of lithium inventory (LLI), because of parasitic reactions, such as the formation of the SEI, may be a reasonable explanation of the aging process before aging cycle 300. After aging cycle 300, it can be discerned from Fig. 7 that the capacity degradation accelerates for the two batteries. There may be some other degradation mechanisms provoking capacity loss, besides loss of lithium inventory beyond aging cycle 300, which can be illustrated by Fig. 8. Between aging cycle 300 and 700, the intensity decrease of peak ② * ② and ③ * ② shows that a less amount of active material is involved in the reac-

tion, and thus loss of active material (LAM) has accelerated the capacity fade ratio after cycle 300. For the battery II (optimal charging cycle aging), the IC curves evolutions of the battery II is similar to the battery I, such that the two batteries with different charging regimes suffer from similar degradation, substantiating that the proposed optimal charging protocol did not reduce battery useful life.

The contributions of various attributes to the capacity degradation in the two life-cycle experiments are shown in Fig. 9, from which degradation inconsistency of the two batteries under different charging regimes can be observed. The contributions from LLI, LAM, and kinetic hindrance to capacity loss are all quantified, as concluded from the test data of charge-discharge cycles at different rates [33]. For the two batteries, there are two degradation stages under the cycle aging. Before aging cycle 300 (stage one), the capacity loss is likely associated with LLI at a rate of 0.021% per cycle for the battery I and 0.027% per cycle for the battery II. The LLI rate of 0.5C CC-CV is slower than the optimal charging, which may be attributed to the fact that the charging current at some SOC range in the optimal charging is much higher. After aging cycle 300 (stage two), except for LLI, LAM also has been an important contribution to capacity fade, while the battery with 0.5C CC-CV charging induced more losses of active material, in comparison with the optimal charging. This is probably because that the optimal charging current is lower after 80% SOC. The kinetic property becomes diminishing at the end of charging, since lithium ion intercalation in graphite negative electrode is more difficult. From Fig. 9, it is evident that the kinetic hindrance of 0.5C CC-CV charging is higher than that in the optimal charging, along with cycle aging. Therefore, the optimal charging facilitates reducing kinetic hindrance.

5. Conclusions

This article first introduces an enhanced thermal behavior model for lithium-ion batteries, in which the function relating polarization voltage to charge current is developed. It is proved that the established thermal model can improve the solution accuracy at high charging current. The optimal charging strategy is devised to balance charging time and temperature rise, with polarization constraints fulfilled. Genetic algorithm is employed to optimize the charging current profile. The experimental results indicate that the proposed charging protocol can reduce charging time remarkably with reasonable temperature rise, highlighting its advantages over conventional CC-CV methods. Aging experiments further verify that the proposed charging protocol has a similar capacity retention ratio, compared to that of 0.5C CC-CV charging after 700 cycles.

Acknowledgements

This work was in part supported by the National Natural Science Foundation of China (Project No. 51277010) and in part by the Fundamental Research Funds for the Central Universities (Project No. 0903005203398).

References

- [1] International Energy Agency (IEA). Technology roadmap: electric and plug-in hybrid electric vehicles Technical report. Paris (France): IEA; 2011.
- [2] Vo TT, Chen XP, Shen WX, Kapoor A. New charging strategy for lithium-ion batteries based on the integration of Taguchi method and state of charge estimation. *J Power Sources* 2015;271:413–22.
- [3] Guo Z, Liaw BY, Qiu XP, Gao L, Zhang CS. Optimal charging method for lithium ion batteries using a universal voltage protocol accommodating aging. *J Power Sources* 2015;274:957–64.

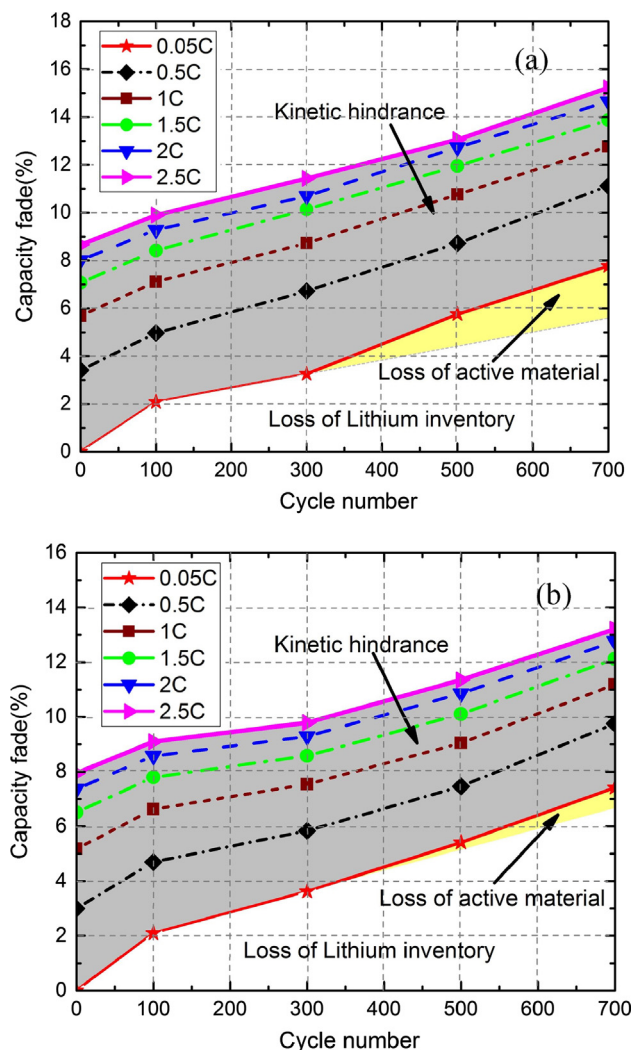


Fig. 9. Capacity fade map describing contributions to the total capacity loss: (a) 0.5C CC-CV cycle aging and (b) optimal charging cycle aging.

- [4] Hu XS, Li S, Peng H, Sun FC. Charging time and loss optimization for LiNMC and LiFePO₄ batteries based on equivalent circuit models. *J Power Sources* 2013;239:449–57.
- [5] Zhang S, Zhang CN, Xiong R, Zhou W. Study on the optimal charging strategy for lithium-ion batteries used in electric vehicles. *Energies* 2014;7:6783–97.
- [6] Hussein Ala A, Fardoun Abbas A, Stephen Samantha S. An online frequency tracking algorithm using terminal voltage spectroscopy for battery optimal charging. *IEEE Trans Sustain Energy* 2016;7(1):32–40.
- [7] Ji Liu, Guang Li, Hosam K, Fathy. A computationally efficient approach for optimizing lithium-ion battery charging. *J Dyn Syst Meas Contr* 2016;138(2).
- [8] Liu YH, Hsieh CH, Luo YF. Search for an optimal five-step charging pattern for Li-ion batteries using consecutive orthogonal arrays. *IEEE Trans Energy Convers* 2011;26(2):654–61.
- [9] Liu YH, Luo YF. Search for an optimal rapid-charging pattern for Li-ion batteries using the Taguchi approach. *IEEE Trans Ind Electron* 2010;57(12):3963–71.
- [10] Anseán D, González M, Viera JC, et al. Fast charging technique for high power lithium iron phosphate batteries: a cycle life analysis. *J Power Sources* 2013;239:9–15.
- [11] Legrand N, Knosp B, Desprez P, et al. Physical characterization of the charging process of a Li-ion battery and prediction of Li plating by electrochemical modelling. *J Power Sources* 2014;245:208–16.
- [12] Chen LR, Wu SL, Shieh DT, et al. Sinusoidal-ripple-current charging strategy and optimal charging frequency study for Li-ion batteries. *IEEE Trans Ind Electron* 2013;60(1):88–97.
- [13] Huang SJ, Huang BG, Pai FS. Fast charge strategy based on the characterization and evaluation of LiFePO₄ batteries. *IEEE Trans Power Electron* 2013;28(4):1555–62.
- [14] Yan JY, Xu GQ, Qian HH, et al. Model predictive control-based fast charging for vehicular batteries. *Energies* 2011;4:1178–96.
- [15] Sourav Pramanik, Sohail Anwar. Electrochemical model based charge optimization for lithium-ion batteries. *J Power Sources* 2016;313:164–77.
- [16] Abdollahi A, Han X, Avvari GV, Raghunathan N, et al. Optimal battery charging, Part I: Minimizing time-to-charge, energy loss, and temperature rise for OCV-resistance battery model. *J Power Sources* 2016;303:388–98.
- [17] Jiang JC, Zhang CP, Wen JP, et al. An optimal charging method for Li-ion batteries using a fuzzy-control approach based on polarization properties. *IEEE Trans Veh Technol* 2013;62(7):3000–9.
- [18] Jiang JC, Liu QJ, Zhang CP, et al. Evaluation of acceptable charging current of power Li-ion batteries based on polarization characteristics. *IEEE Trans Ind Electron* 2014;61(12):6844–51.
- [19] Kim KM, Lee YG, Kang KY, et al. A simple method for solving the voltage overshoots of LiFePO₄-based lithium-ion batteries with different capacity classes. *RSC Adv* 2012;2(9):3844–9.
- [20] Xiong R, Sun F, Chen Z, He H. A data-driven multi-scale extended Kalman filtering based parameter and state estimation approach of lithium-ion polymer battery in electric vehicles. *Appl Energy* 2014;113:463–76.
- [21] Sun F, Xiong R. A novel dual-scale cell state-of-charge estimation approach for series-connected battery pack used in electric vehicles. *J Power Sources* 2015;274:582–94.
- [22] Sun F, Xiong R, He H. A systematic state-of-charge estimation framework for multi-cell battery pack in electric vehicles using bias correction technique. *Appl Energy* 2016;162:1399–409.
- [23] Hu XS, Li SB, Peng H. A comparative study of equivalent circuit models for Li-ion batteries. *J Power Sources* 2012;198:359–67.
- [24] Zhang CP, Wang LY, Li X, Chen W, Yin GG, Jiang JC. Robust and adaptive estimation of state of charge for lithium-ion batteries. *IEEE Trans Ind Electron* 2015;62(8):4948–57.
- [25] Liao LX, Zuo PJ, Ma YL, et al. Effects of temperature on charge/discharge behaviors of LiFePO₄ cathode for Li-ion batteries. *Electrochim Acta* 2012;60:269–73.
- [26] Dubarry M, Truchot C, Liaw BY, et al. Evaluation of commercial lithium-ion cells based on composite positive electrode for plug-in hybrid electric vehicle applications III. Effect of thermal excursions without prolonged thermal aging. *J Electrochem Soc* 2013;160(1):A191–9.
- [27] Leng F, Tan CM, Pecht M. Effect of temperature on the aging rate of Li ion battery operating above room temperature. *Sci Rep* 2015;5:1–12.
- [28] Ma ZY, Jiang JC, Shi W, et al. Investigation of path dependence in commercial lithium-ion cells for pure electric bus applications: aging mechanism identification. *J Power Sources* 2015;274:29–40.
- [29] Onda K, Ohshima T, Nakayama M, et al. Thermal behavior of small lithium-ion battery during rapid charge and discharge cycles. *J Power Sources* 2006;158:535–42.
- [30] Sajan KS, Kumar V, Tyagi B. Genetic algorithm based support vector machine for on-line voltage stability monitoring. *Int J Electr Power Energy Syst* 2015;73:200–8.
- [31] Gholami R, Shahabi M, Haghifam MR. An efficient optimal capacitor allocation in DG embedded distribution networks with islanding operation capability of micro-grid using a new genetic based algorithm. *Int J Electr Power Energy Syst* 2015;71:335–43.
- [32] Dubarry M, Truchot C, Liaw BY. Synthesize battery degradation modes via a diagnostic and prognostic model. *J Power Sources* 2012;219:204–16.
- [33] Dubarry M, Truchot C, Liaw BY, et al. Evaluation of commercial lithium-ion cells based on composite positive electrode for plug-in hybrid electric vehicle applications. Part II. Degradation mechanism under 2 C cycle aging. *J Power Sources* 2011;196:10336–43.